



1



2

Contents

- Discriminative models vs Conditional generative models vs unconditional generative models
- Conditional GAN
- Conditional VAE -
- Conditional diffusional model (Stable diffusion)



3

Discriminative vs Generative Models

Discriminative Model:

Learn a probability distribution $p(y|x)$

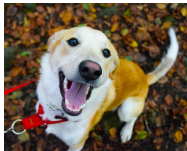


$$P(\text{cat} | \text{image}) = 80\%$$

$$P(\text{dog} | \text{image}) = 20\%$$

Generative Model:

Learn a probability distribution $p(x)$



$$P(\text{cat} | \text{image}) = 20\%$$

$$P(\text{dog} | \text{image}) = 80\%$$

Conditional Generative Model: Learn $p(x|y)$

Discriminative model: the possible labels for each input "compete" for probability mass.
But no competition between **images**



4

Discriminative vs Generative Models

Discriminative Model:
Learn a probability
distribution $p(y|x)$



$$P(\text{cat} | \text{img}) = 60\%$$

$$P(\text{dog} | \text{img}) = 40\%$$

Generative Model:
Learn a probability
distribution $p(x)$



$$P(\text{cat} | \text{img}) = 20\%$$

$$P(\text{dog} | \text{img}) = 80\%$$

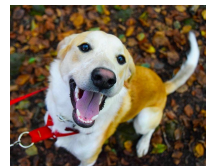
Conditional Generative Model: Learn $p(x|y)$

Discriminative model: No way for the model to handle unreasonable inputs; it must give label distributions for all images



Discriminative vs Generative Models

Discriminative Model:
Learn a probability
distribution $p(y|x)$



$$P(\text{img})$$

$$P(\text{img})$$

$$P(\text{img})$$

Generative Model:
Learn a probability
distribution $p(x)$

Conditional Generative Model: Learn $p(x|y)$

Generative model: All possible images compete with each other for probability mass
Requires deep image understanding! Is a dog more likely to sit or stand? How about 3-legged dog vs 3-armed monkey?
Model can “reject” unreasonable inputs by assigning them small values



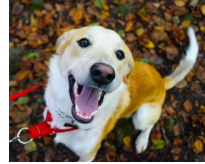
Discriminative vs Generative Models

Discriminative Model:

Learn a probability distribution $p(y|x)$



$P(\text{cat} | \text{cat})$



$P(\text{cat} | \text{dog})$



$P(\text{cat} | \text{monkey})$

Generative Model:

Learn a probability distribution $p(x)$

$P(\text{dog} | \text{cat})$

$P(\text{dog} | \text{dog})$

$P(\text{dog} | \text{monkey})$

Conditional Generative Model: Learn $p(x|y)$

Conditional Generative Model: Each possible label induces a competition among all images

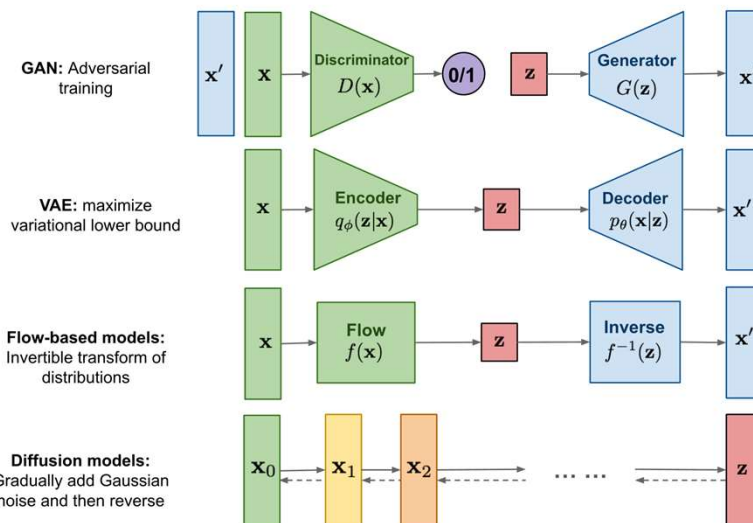


CV Group – School of Information and Communication and Technology

7

7

Comparing Generative Models



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

8

CONDITIONAL GAN



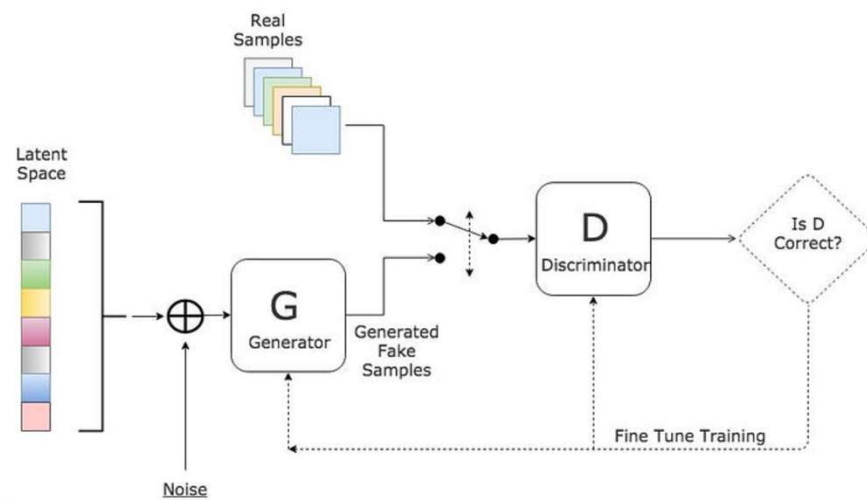
ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

CV Group – School of Information and Communication and Technology

9

9

Generative Adversarial Network



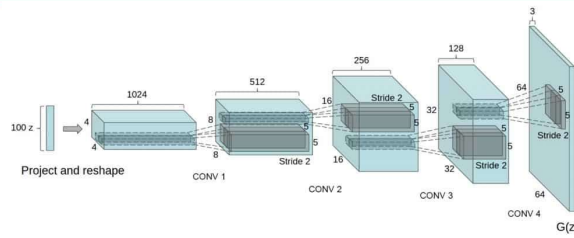
ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

CV Group – School of Information and Communication and Technology

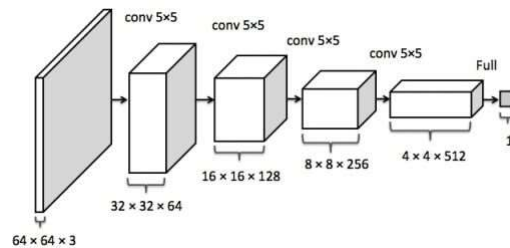
10

10

Deep Convolutional GAN (DCGAN)



Generator of DCGAN with fractionally strided convolutional layers



Discriminator of DCGAN with strided convolutional layers



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

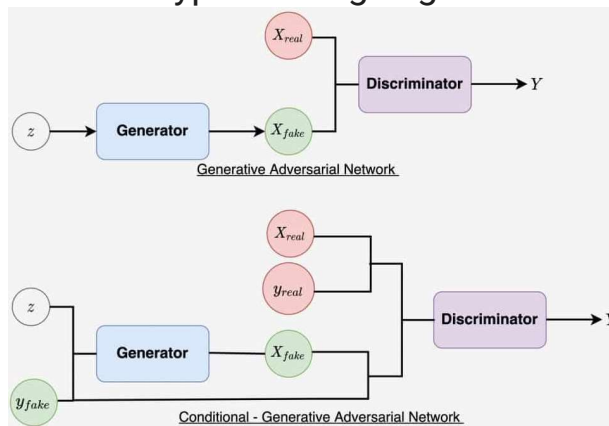
CV Group – School of Information and Communication and Technology

11

11

GAN problem

- Though the GAN model can generate new realistic samples for a particular dataset, we have zero control over the type of images generated.



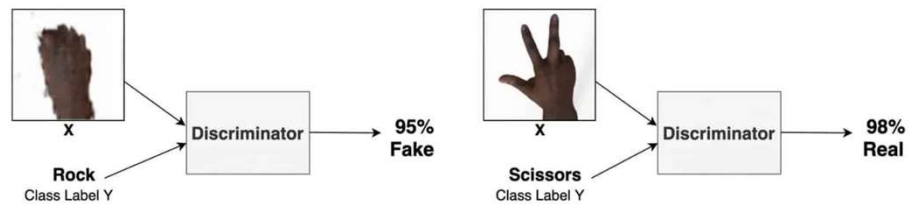
ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

CV Group – School of Information and Communication and Technology

12

12

Discriminator of Conditional DCGAN



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

CV Group – School of Information and Communication and Technology

13

13

CONDITIONAL VARIATIONAL AUTOENCODERS



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

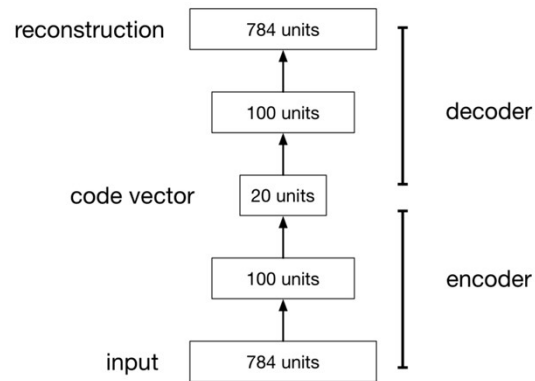
CV Group – School of Information and Communication and Technology

14

14

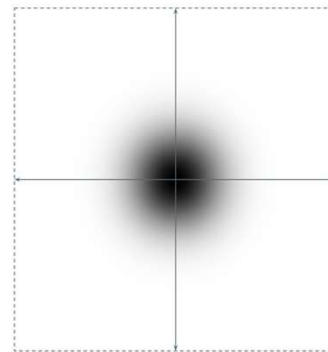
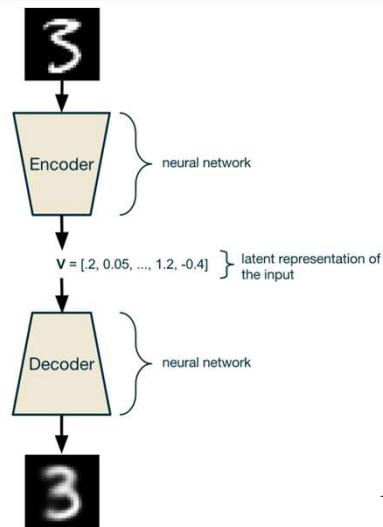
Autoencoders

- An autoencoder is a feed-forward neural net whose job it is to take an input x and predict x .
- To make this non-trivial, we need to add a bottleneck layer whose dimension is much smaller than the input.



15

Autoencoders (2)



A standard normal distribution in latent space

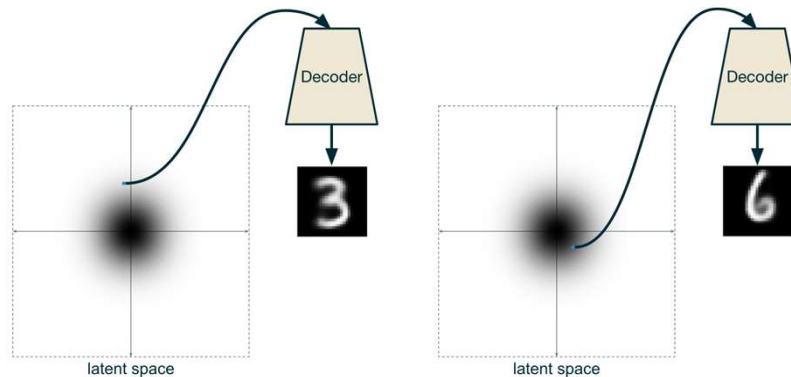
A standard autoencoder



16

The encoder of a variational autoencoder

- We would like to train the decoder to take any point sampled from this distribution and return a reasonable digit image.



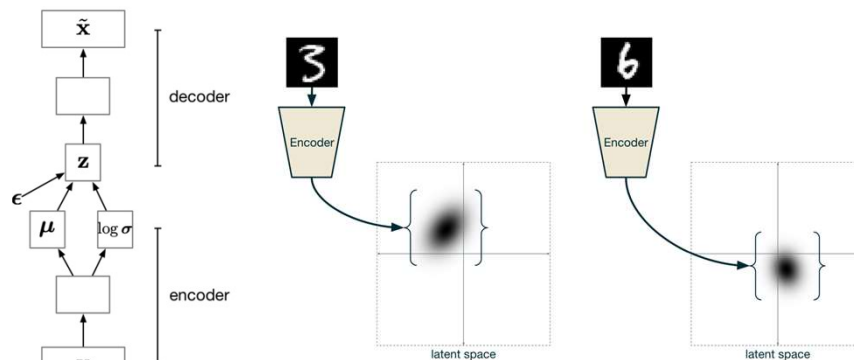
CV Group – School of Information and Communication and Technology

17

17

The encoder of a variational autoencoder

- In a variational autoencoder, the encoder instead produces a probability distribution in the latent space.
- The latent distributions it outputs are gaussians of the same dimensionality as the latent space. The encoder produces the parameters of these gaussians.

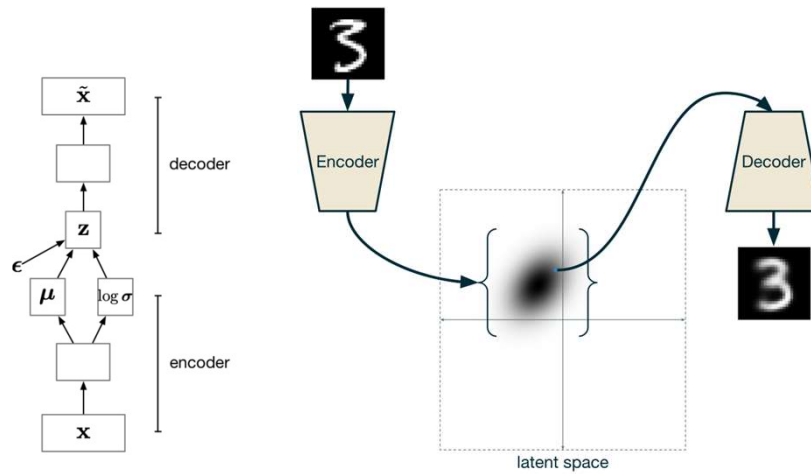


CV Group – School of Information and Communication and Technology

18

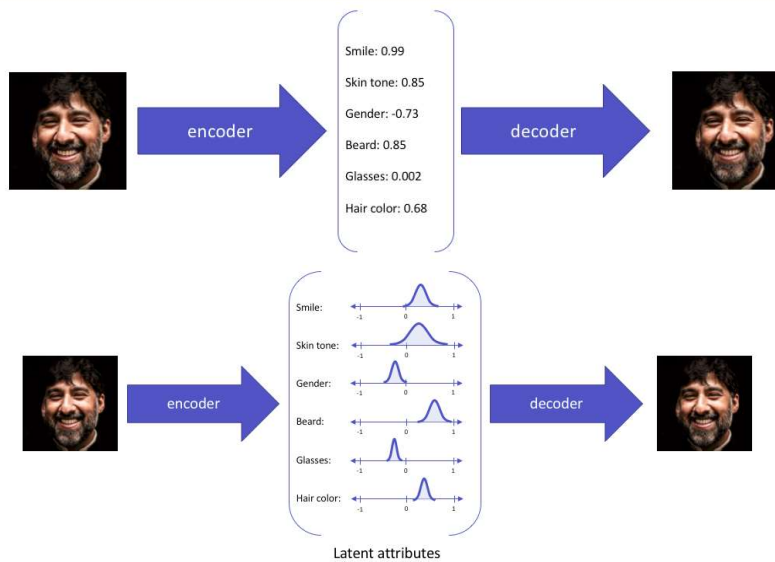
18

A variational autoencoders



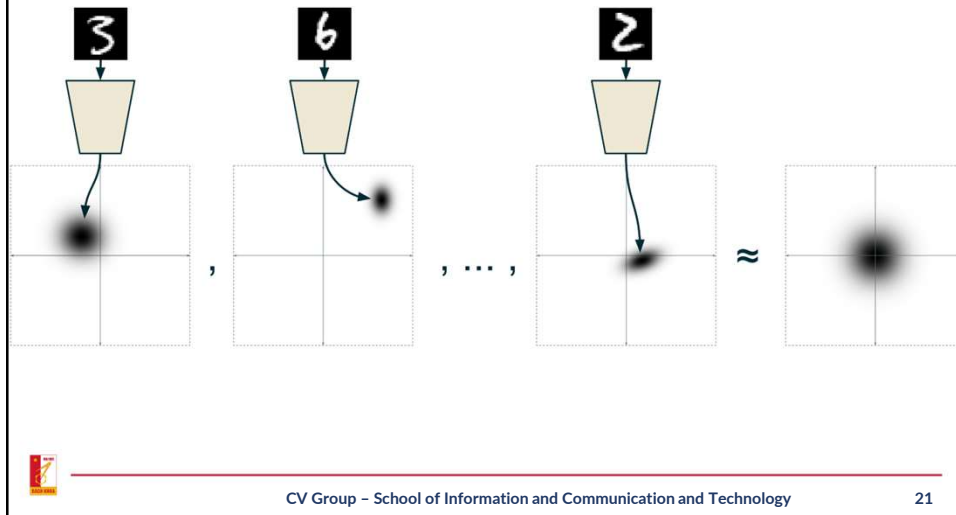
19

Differences between autoencoders and variational autoencoders



20

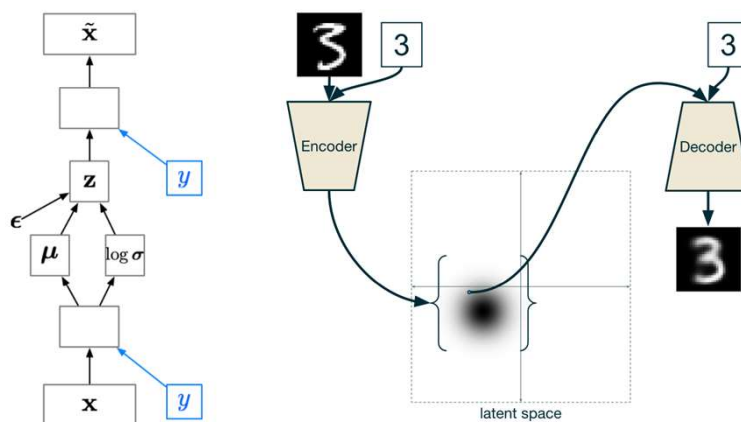
The Structure of the Latent Space



21

Conditional Variational Autoencoder

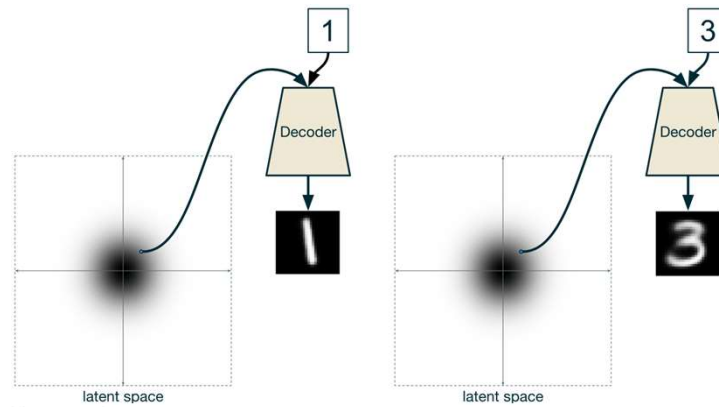
- An extra input to both the encoder and the decoder at training time



22

Conditional Variational Autoencoder (2)

- To generate an image of a particular number, just feed that number into the decoder along with a random point in the latent space sampled from a standard normal distribution.



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

CV Group – School of Information and Communication and Technology

23

23

CONDITIONAL DIFFUSION MODEL (STABLE DIFFUSION MODEL)

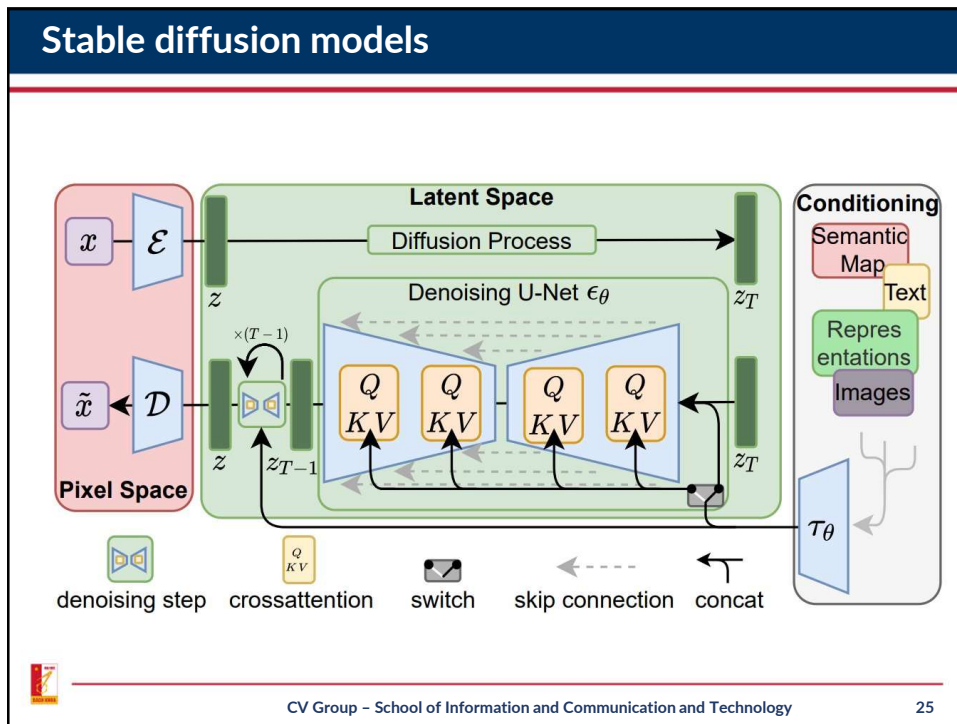


ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

CV Group – School of Information and Communication and Technology

24

24



25

Diffusion models

- Forward diffusion (noising)
 - $x_0 \rightarrow x_1 \rightarrow \dots x_T$
 - Take a data distribution $x_0 \sim p(x)$, turn it into noise by diffusion $x_T \sim \mathcal{N}(0, \sigma^2 I)$

- Reverse diffusion (denoising)
 - $x_T \rightarrow x_{T-1} \rightarrow \dots x_0$
 - Sample from the noise distribution $x_T \sim \mathcal{N}(0, \sigma^2 I)$, reverse the diffusion process to generate data $x_0 \sim p(x)$

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

26

Math Formalism

- For a forward diffusion process

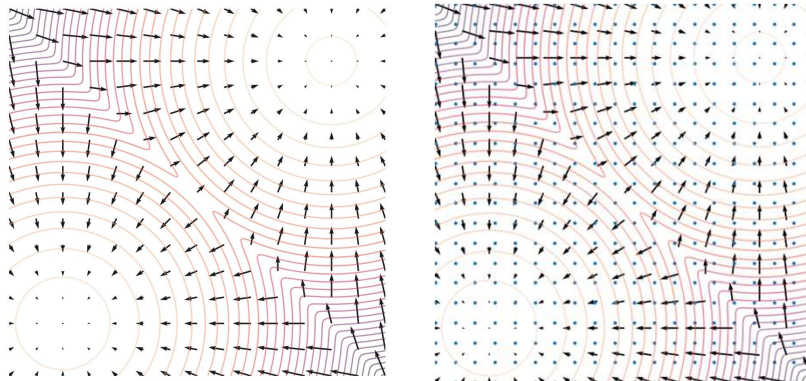
$$d\mathbf{x} = f(\mathbf{x}, t)dt + g(t)d\mathbf{w}$$

- There is a backward diffusion process that reverse the time

$$d\mathbf{x} = [f(\mathbf{x}, t) - g(t)^2 \nabla_{\mathbf{x}} \log p(\mathbf{x}, t)]dt + g(t)d\mathbf{w}$$

- If we know the *time-dependent* score function $\nabla_{\mathbf{x}} \log p(\mathbf{x}, t)$
- Then we can reverse the diffusion process.

Animation for the Reverse Diffusion



<https://yang-song.net/blog/2021/score/>

Training diffusion model = Learning to denoise

- If we can learn a score model

$$f_{\theta}(x, t) \approx \nabla \log p(x, t)$$
 - Then we can denoise samples, by running the reverse diffusion equation. $x_t \rightarrow x_{t-1}$
- Score model $f_{\theta}: \mathcal{X} \times [0, 1] \rightarrow \mathcal{X}$
 - A time dependent vector field over x space.
- Training objective: Infer noise from a noised sample

$$x \sim p(x), \epsilon \sim \mathcal{N}(0, I), t \in [0, 1]$$

$$\min \|\epsilon + f_{\theta}(x + \sigma^t \epsilon, t)\|_2^2$$
 - Add Gaussian noise ϵ to an image x with scale σ^t , learn to infer the noise σ .



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

29

Conditional denoising

- Infer noise from a noised sample, based on a condition y
 - $x, y \sim p(x, y), \epsilon \sim \mathcal{N}(0, I), t \in [0, 1]$
 - $\min \|\epsilon - f_{\theta}(x + \sigma^t \epsilon, y, t)\|_2^2$
- Conditional score model $f_{\theta}: \mathcal{X} \times \mathcal{Y} \times [0, 1] \rightarrow \mathcal{X}$
 - Use Unet as to model image to image mapping
 - Modulate the Unet with condition (text prompt).

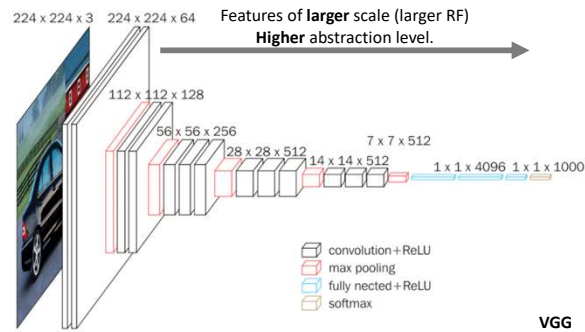


ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

30

Convolutional Neural Network

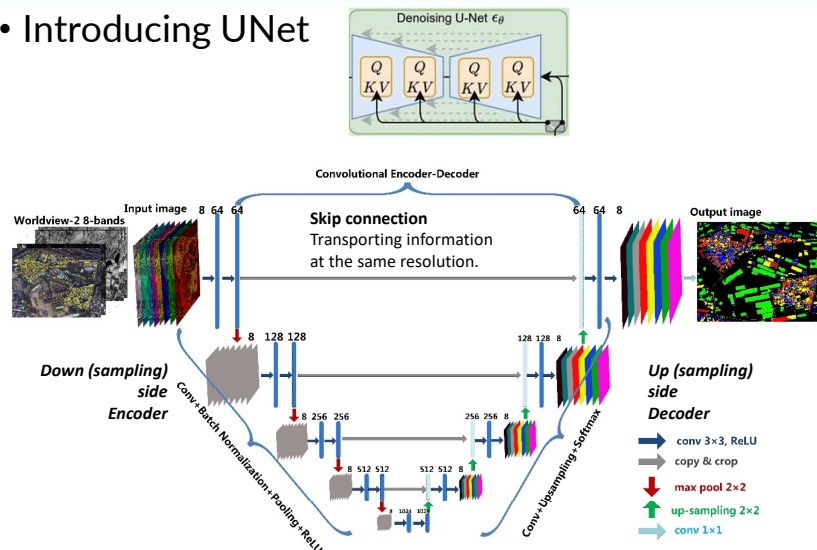
- CNN parametrizes function over images
- Motivation
 - Features are translational invariant
 - Extract feature at different scale / abstraction level
- Key modules
 - Convolution
 - Downsampling (Max-pool)



31

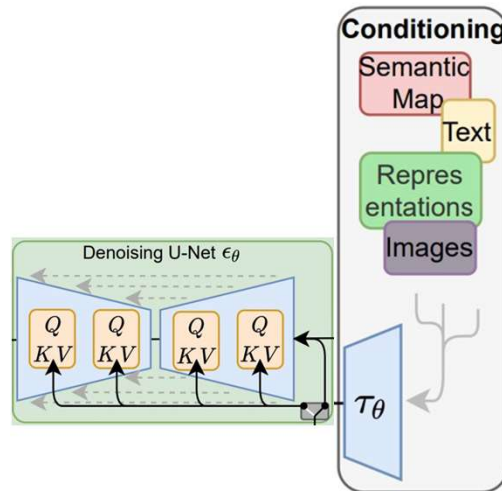
UNet: a natural architecture for image-to-image function

- Introducing UNet



32

How to understand prompts?



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

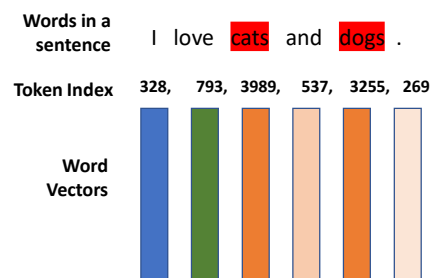
CV Group – School of Information and Communication and Technology

33

33

Word as Vectors: Language Model 101

- Unlike pixel, meaning of word are not explicitly in the characters.
- Word can be represented as index in dictionary
 - But index is also meaning less.
- Represent words in a vector space
 - Vector geometry => semantic relation.

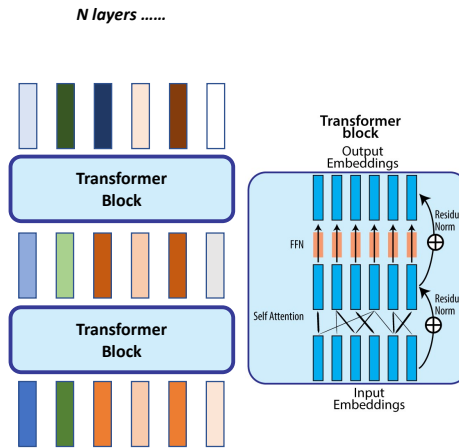


ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

34

Word Vector in Context: DNN / Transformers

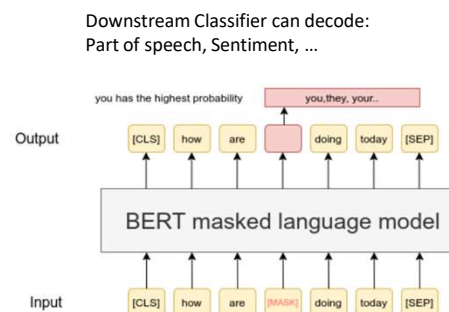
- Meaning of word depends on context, not always the same.
 - “I book a ticket to buy that book.”
- Word vectors should depend on context.
- Transformers let each word “absorb” influence from other words to be “contextualized”



35

Learning Word Vectors: GPT & BERT & CLIP

- Self-supervised learning of word representation
 - Predicting missing / next words in a sentence. (BERT, GPT)
 - Contrastive Learning, matching image and text. (CLIP)

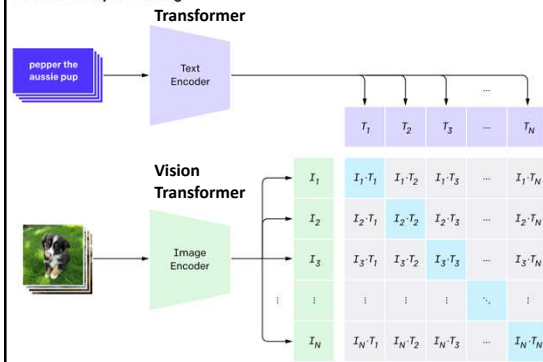


[MLM — Sentence-Transformers documentation \(sbert.net\)](https://huggingface.co/docs/sentence-transformers/sentence-transformers-documentation)

36

Joint Representation for Vision and Language : CLIP

1. Contrastive pre-training

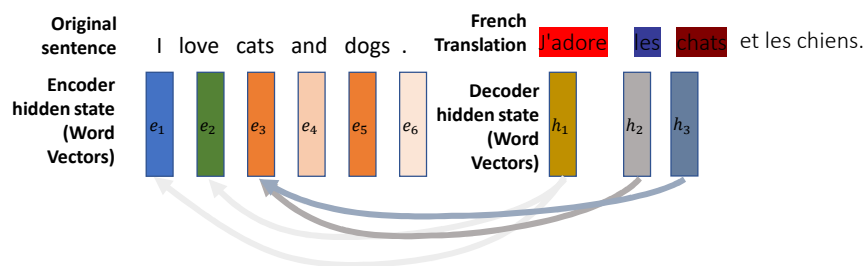


- Learn a joint encoding space for text caption and image
- Maximize representation similarity between an image and its caption.
- Minimize other pairs

37

How does text affect diffusion?

- Origin of Attention: Machine Translation (Seq2Seq)

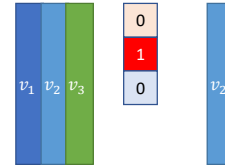


- Use **Attention** to retrieve useful info from a batch of vectors.

38

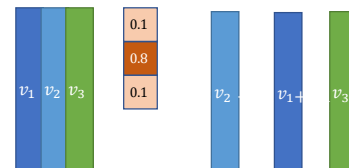
From Dictionary to Attention - Dictionary: Hard-indexing

- $\text{dic} = \{1 : v_1, 2 : v_2, 3 : v_3\}$
 - Keys 1,2,3
 - Values v_1, v_2, v_3
- $\text{dic}[2]$
 - Query 2
 - Find 2 in keys
 - Get corresponding value.
- Retrieving values as matrix vector product
 - One hot vector over the keys
 - Matrix vector product



From Dictionary to Attention - Attention: Soft-indexing

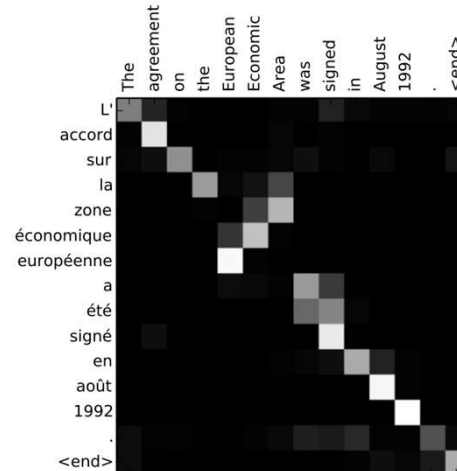
- Soft indexing
 - Define an attention distribution a over the keys
- Matrix vector product.
- Distribution based on similarity of query and key.



Visualizing Attention matrix a_{ij}

- French 2 English
- “Learnt to pay Attention”
 - “la zone économique européenne” -> “the European Economic Area”
 - “a été signé” -> “was signed”

<https://jalammar.github.io/visualizing-neural-machine-translation-mechanics-of-seq2seq-models-with-attention/>

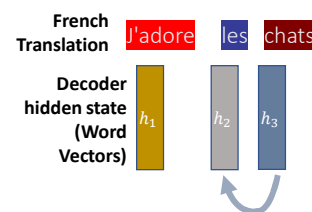


ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

41

Cross & Self Attention

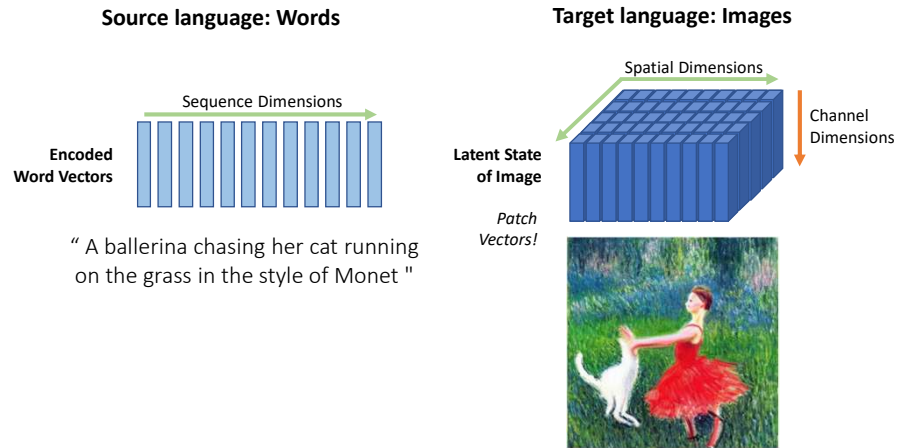
- Cross Attention
 - Tokens in one language pay attention to tokens in **another**.
- Self Attention ($e_i = h_i$)
 - Tokens in a language pay attention to **each other**.



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

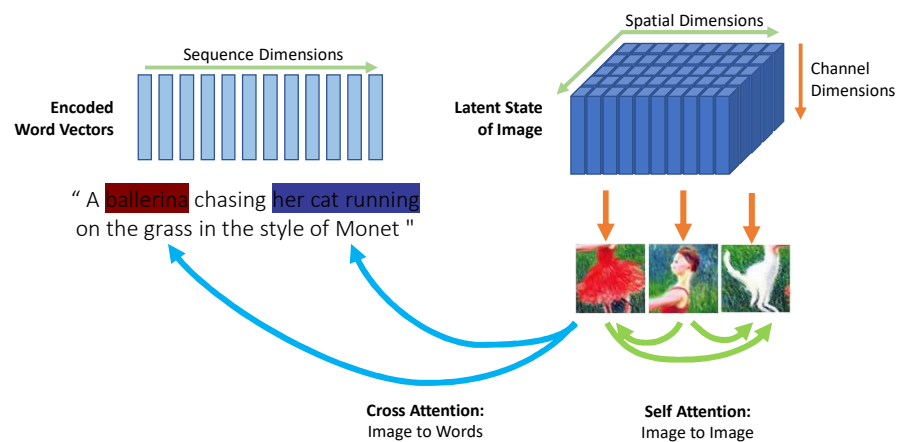
42

Text2Image as translation



43

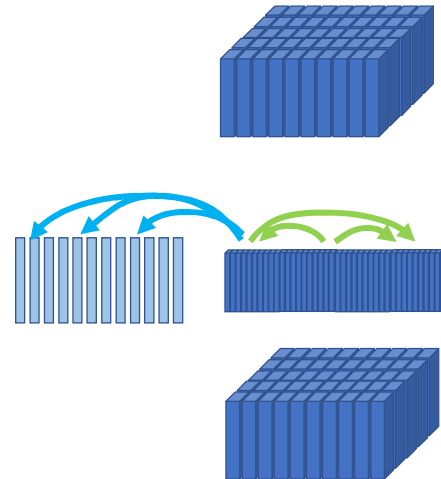
Text2Image as translation



44

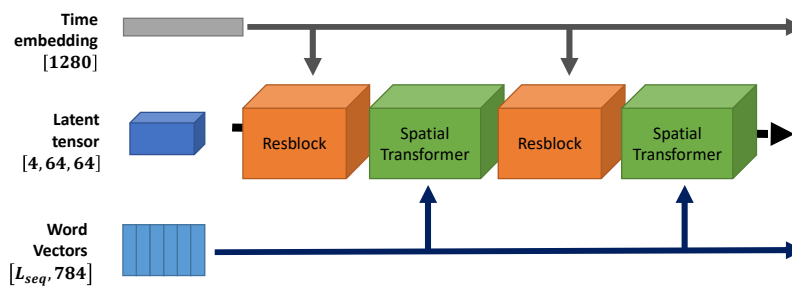
Spatial Transformer

- Rearrange spatial tensor to sequence.
- Cross Attention
- Self Attention
- FFN
- Rearrange back to spatial tensor (same shape)



45

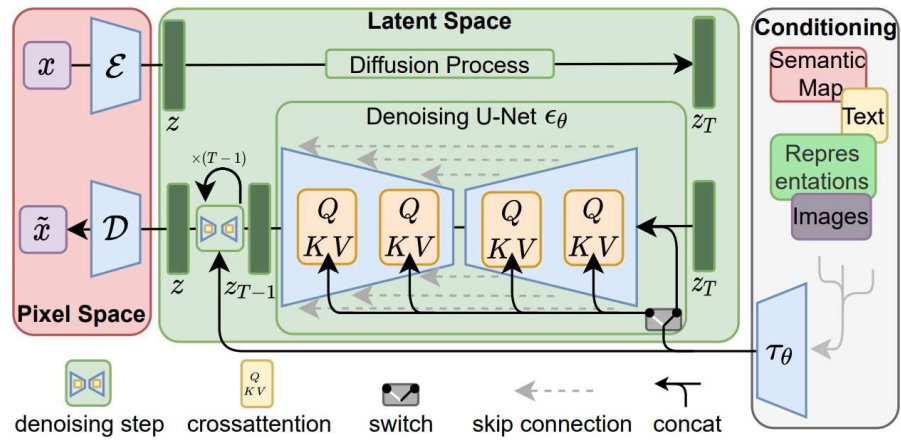
Spatial transformer + ResBlock (Conv layer)



- Alternating Time and Word Modulation
- Alternating Local and Nonlocal operation

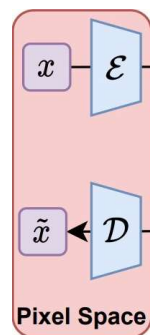
46

Diffusion in Latent Space



47

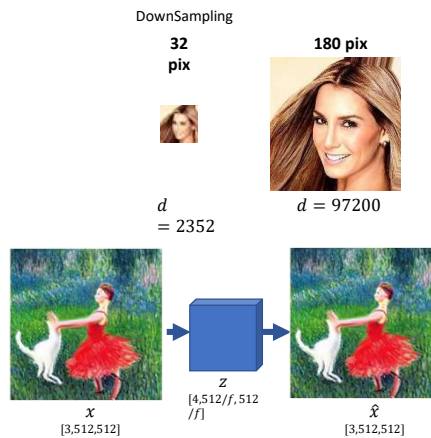
Diffusion in Latent Space



48

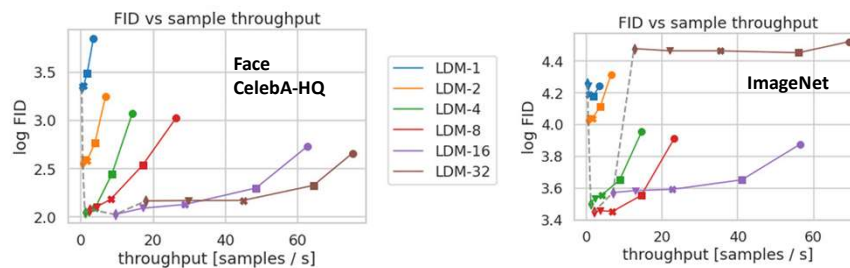
Diffusion in latent space

- Motivation:
 - Natural images are high dimensional
 - but have many redundant details that could be compressed / statistically filled out
- Division of labor
 - Diffusion model -> Generate low resolution sketch
 - AutoEncoder -> Fill out high resolution details
- Train a VAE model to compress images into latent space.
 - $x \rightarrow z \rightarrow \hat{x}$
- Train diffusion models in latent space of z .



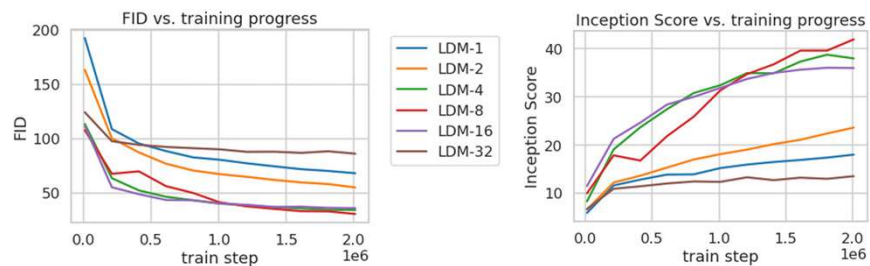
Spatial Compression Tradeoff

- LDM- $\{f\}$. f = Spatial downsampling factor
 - Higher f leads to faster sampling, with degraded image quality (FID $\uparrow$)
 - Fewer sampling steps leads to faster sampling, with lower quality (FID $\uparrow$)



Spatial Compression Tradeoff

- LDM- $\{f\}$. f = Spatial downsampling factor
 - Too little compression $f = 1, 2$ or too much compression $f = 32$, makes diffusion hard to train.



51

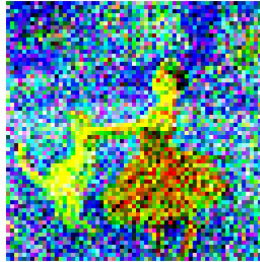
Details in Stable Diffusion

- In stable diffusion, spatial downsampling $f = 8$
 - x is (3, 512, 512) image tensor
 - z is (4, 64, 64) latent tensor
- Regularizing the Latent Space
 - KL regularizer
 - Similar to VAE, make latent distribution like Gaussian distribution.
 - VQ regularizer
 - Make the latent representation quantized to be a set of discrete tokens.

52

Meaning of latent space

- Latent state contains a “sketch version” of the image.



$z[0:3,:,:]$



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

53

Diffusion Process Visualized



ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

54

